A simplified unified propagator model for signed order flow, concave price impact and rough volatility

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Abstract

We simplify the setup in [MORS26] by modelling persistent signed order flow F_t as the difference of two integrated hyper-rough Heston-type processes with an L^1-kernel and Hölder exponent in $(\frac{1}{2},1)$, which endogenously determines the (pure feedback) order-driven martingale price process P, the price impact kernel $G(.)$ for the propagator equation (which is independent of the vol-of-vol parameter ν), and the celebrated square-root impact law if we let $H \to 0$. The persistence explains the transient component of the price impact, and our approach obviates the need to use nearly unstable heavy-tailed Hawkes processes, and lengthy macroscopic scaling limit arguments with C-tightness as in [MORS26]. We also discuss longer-memory/multiscale behaviour for F, and the addition of proportional bid-ask spreads and exogenous choice of price impact kernel, wherein P is no longer a martingale but the model is still arbitrage-free if P satisfies the conditional full support property in Guasoni et al.[GRS08]. We also propose alternate models where signed order flow is a Gaussian Volterra process with $H \in (\frac{1}{2},1)$, or using fGN which leads to power-law decay for the a.c.f. of the order flow sign process as in [LMF05],[MB25].

1.1 Persistent order flow

Following [AJ21] (and similar to Theorem 3.1 in [MORS26]), we model the cumulative positive order flow F_t^+ for an asset (i.e. the total number of shares bought by time t) as a weak solution to the Stochastic Volterra Equation:

$$F_t^+ \quad = \quad g(t) + \int_0^t f(t-s)M_s ds \tag{1}$$

where M is a continuous local martingale with $M_0 = 0$ and $[M]_t = F_t^+$ for $t \geq 0$, $f(t) = \nu e^{-\theta t} t^{H-\frac{1}{2}}$ with $H \in (-\frac{1}{4},0)$, $\nu, \theta > 0$ so $f \in L^1$, and $g(t) = V_0 t$ for some $V_0 > 0$ (note θ replaces the ergodicity parameter λ in Theorem 3.1 in [MORS26]). Weak existence and uniqueness for (1) is established in Theorem 2.2 of [AJ21] in terms of a Volterra Integral Equation (VIE), and F^+ is a non-decreasing, nonnegative, continuous and adapted process. F_t^+ is also a.s. $(2\alpha - \varepsilon) \wedge 1$-Hölder continuous where $\alpha = H + \frac{1}{2}$ (see e.g. Theorem 3.1 in [JR20]), so from the condition on H we see that $2\alpha \in (\frac{1}{2},1)$ [1], and F_t^+ is non-differentiable if $H \leq 0$. Note that if $H > 0$, F^+ is just the integrated variance $\int_0^t V_s ds$ for a standard (i.e. non-hyper) Rough-Heston type variance process V, see e.g. section 5.4 of [FGS21] for details.

Lemma 1.1 $\mathbb{E}(F_t^+) < \infty$.

Proof. See Appendix A. ■

Corollary 1.2 *Assume (F^+, M) is a weak solution of (1) in the sense of [AJ21], with $b = 0$, $c = 1$ and $\nu \equiv 0$. Then M is a continuous square-integrable martingale.*

Proof. Just follows from the previous lemma and Proposition 1.1.11 in [Pham09]. ■

1.2 The inversion formula

If h is a function which solves $h * f \equiv 1$, then we have the inversion formula:

$$\begin{aligned}
\int_0^t h(t-s)(F_s^+ - g(s))ds \quad &= \quad \int_0^t h(t-s)\int_0^s f(s-u)M_u du ds \quad = \quad \int_0^t \int_u^t h(t-s)f(s-u)ds\, M_u du \\
&= \quad \int_0^t \int_0^{t-u} h(t-s-u)f(s)ds\, M_u du \\
&= \quad \int_0^t (h*f)(t-u)M_u du \quad = \quad \int_0^t M_u du
\end{aligned}$$

(where Fubini is justified since $F_t^+ < \infty$ a.s. for finite t and M is continuous), and hence

$$M_t \quad = \quad \frac{d}{dt}\int_0^t h(t-s)(F_s^+ - g(s))ds \quad = \quad \frac{d}{dt}\int_0^t h(t-s)F_s^+ ds - \frac{d}{dt}\int_0^t h(t-s)g(s)ds \tag{2}$$

[1] we can simulate F^+ using the Monte Carlo scheme in [AA25] using Normal Inverse Gaussian variates

Lebesgue a.e., where the final equality holds since $\int_0^t h(t-s)(F_s^+ - g(s))ds$ and $\int_0^t h(t-s)g(s)ds$ are both absolutely continuous, and hence so is their sum. Moreover, setting $\bar{H}(t) = \int_0^t h(s)ds$, integrating by parts we see that

$$\int_0^t \bar{H}(t-s)dF_s^+ \;=\; F_s^+\bar{H}(t-s)_{s=0}^{s=t} + \int_0^t h(t-s)F_s^+ ds \;=\; \int_0^t h(t-s)F_s^+ ds\,.$$

Taking Laplace transforms, the condition $h * f \equiv 1$ becomes $\hat{f}(\lambda)\hat{h}(\lambda) = \frac{1}{\lambda}$, from which we find that

$$h(t) \;=\; \frac{\theta^\alpha}{\nu\Gamma(\alpha)}(1 - \frac{\Gamma(-\alpha, t\theta)}{\Gamma(-\alpha)})$$

where $\Gamma(a,z) = \int_z^\infty s^{a-1}e^{-s}ds$ is the (analytic continuation) of the incomplete Gamma function, and $h(t) = O(t^{-H-\frac{1}{2}}) \to \infty$ as $t \to 0$.

Remark 1.1 Note that $\Gamma(-\alpha, t\theta)$ tend to $+\infty$ not $\Gamma(-\alpha)$ as $t \to 0$ since $-\alpha < 0$.

Moreover (from the Final Value Theorem for Laplace transforms), we see that

$$h(\infty) \;=\; \lim_{t\to\infty} h(t) \;=\; \lim_{\lambda\to 0} \lambda\hat{h}(\lambda) \;=\; \frac{1}{\hat{f}(0)} \;=\; \frac{1}{\|f\|_{L^1}} \;=\; \frac{\theta^\alpha}{\nu\Gamma(\alpha)}\,. \tag{3}$$

1.3 Signed order flow

Again following [MORS26], we now assume the *signed* order flow for an asset is

$$F_t \;=\; F_t^+ - F_t^-$$

where $F_t^- = g(t) + \int_0^t f(t-s)M_s^- ds$ and (M^-, F^-) is an independent copy of (M, F^+). Then we see that

$$\frac{d}{dt}\int_0^t h(t-s)F_s ds \;=\; \frac{d}{dt}\int_0^t \bar{H}(t-s)dF_s \;=\; \int_0^t h(t-s)dF_s \;=\; M_t - M_t^- \tag{4}$$

since the g terms cancel (and the second equality follows from the Stieltjes-Leibniz rule using the known Hölder continuity of $F^\pm$ and that $|h(u)| \le Cu^{-\alpha}$ so $\int_0^t |h(t-s)|(dF_s^+ + dF_s^-) < \infty$), and for $0 \le t \le u$ we have

$$\mathbb{E}(F_u^+|\mathcal{F}_t^M) \;=\; g(u) + \mathbb{E}(\int_0^u f(u-s)M_s ds\,|\mathcal{F}_t^M) \;=\; g(u) + \int_0^t f(u-s)M_s ds + M_t\int_t^u f(u-s)ds$$

(and similarly for $\mathbb{E}(F_u^-|\mathcal{F}_t^{M^-})$), since $(M_t)_{t\ge 0}$ is an $(\mathcal{F}_t^M)_{t\ge 0}$-martingale by Corollary 1.2, so

$$\mathbb{E}(F_u^+ - F_u^-|\mathcal{G}_t) \;=\; \int_0^t f(u-s)(M_s - M_s^-)ds + (M_t - M_t^-)\int_t^u f(u-s)ds \tag{5}$$

where $\mathcal{G}_t = \sigma(\mathcal{F}_t^M, \mathcal{F}_t^{M^-})$.

1.4 Endogenous order-driven price dynamics and the propagator equation

We now make the common assumption (cf.[JR20]) that the asset price $P_t = \kappa \lim_{u\to\infty} \mathbb{E}(F_u|\mathcal{G}_t)$ where $F_t = F_t^+ - F_t^-$ (for some $\kappa > 0$) [2]. Then letting $u \to \infty$ in (5), we find that $f(u-s) \searrow 0$ as $u \to \infty$ since $\theta > 0$ so the first term in (5) goes to zero since $|\int_0^t f(u-s)(M_s - M_s^-)ds| \le \sup_{s\in[0,t]} |M_s - M_s^-| \int_0^t f(u-s)ds$ using the monotone convergence theorem, but $\lim_{u\to\infty}\int_t^u f(u-s)ds = \lim_{u\to\infty}\int_0^{u-t} f(r)dr = \int_0^\infty f(r)dr = \|f\|_{L^1} = 1/h(\infty)$ (from (3)), so

$$P_t \;=\; \kappa \lim_{u\to\infty} \mathbb{E}(F_u^+ - F_u^-\,|\mathcal{G}_t) \;=\; \frac{\kappa}{h(\infty)}(M_t - M_t^-) \;=\; \frac{\kappa\nu\Gamma(\alpha)}{\theta^\alpha}(M_t - M_t^-)$$

i.e. P is the difference of two i.i.d. driftless hyper-rough-type Heston models (which effectively both have correlation $\rho = 1$ since M and M^- only have one driving Brownian motion each so P is a "pure feedback" price process, i.e. only depends on its own history). We also recover the propagator equation:

$$P_t \;=\; \int_0^t G(t-s)dF_s$$

where the price impact kernel G is given by

$$G(t) \;=\; \kappa\frac{h(t)}{h(\infty)} \;=\; \kappa(1 - \frac{\Gamma(-\alpha, t\theta)}{\Gamma(-\alpha)}) \tag{6}$$

(recall Remark 1.1) and we note that G is independent of ν. We also define the *unsigned* order flow as $V_t = F_t^+ + F_t^-$.

[2] note we are assuming $P_0 = 0$ without loss of generality to ease notation, but we can easily add a positive P_0 term

1.5 Power-law price impact

In particular, the *market impact function* of an exogenous metaorder executed at constant trading speed 1 up to time t_0 is given by

$$MI(t) \quad = \quad \int_0^{t\wedge t_0} G(t-s)ds \tag{7}$$

(where $\alpha_- = \frac{1}{2} - H$) which is $O(t^{\frac{1}{2}-H})$ as $t \to 0$ (for $H \neq \frac{1}{2}$) and globally concave in t for $H \in (-\frac{1}{2}, \frac{1}{2})$, consistent with empirical evidence (note the usual *square-root law* corresponds to $H = 0$ here). Upon cancelling terms, we can show that the asymptotic (i.e. permanent) impact as $t \to \infty$ (given no trading after t_0) is

$$MI(\infty) \quad = \quad \lim_{t\to\infty} \int_0^{t\wedge t_0} G(t-s)\, ds \quad = \quad \int_0^{t_0} G(\infty)ds \quad = \quad \kappa t_0 \lim_{t\to\infty} \frac{h(t)}{h(\infty)} \quad = \quad \kappa t_0$$

(this corresponds to the asymptotic grey line in Figure 2 below).

For the special case $H = \frac{1}{2}$ that we have previously excluded, $h(t) = \frac{1}{\nu}(\theta + \delta(t))$, which corresponds to linear permanent price impact plus temporary price impact (coming from the Dirac term); conversely $\lim_{\theta\to 0} h(t) = const. \times t^{-\frac{1}{2}-H}$. Recall from above that $(F_t^+)_{t\geq 0}$ has Hölder regularity $(2\alpha - \varepsilon) \wedge 1$, and empirical evidence in [MORS26] and below suggests that $2\alpha \in (\frac{1}{2}, 1)$ which corresponds to our standing assumption that $H \in (-\frac{1}{4}, 0)$.

1.6 Empirical results

Below we have tabulated estimates of 2α (i.e. the Hölder exponent of $(F_t)_{t\geq 0}$) using the signed order flow data in Figure 1 over a 6.5hr trading day for major US tech stocks using the model-independent $\hat{H}^{\pi}_{L,K}(X)$ estimator in Eq 7 in Cont&Das[CD24]. This estimator is consistent and scale-independent and allows for irregularly spaced data (with their $L/K = [\sqrt{N}]$, where N is the total sample size). For this we have used the free data at `https://data.lobsterdata.com/info/DataSamples.php` (note we only use columns 4 and 5 from the lobsterdata.csv files which correspond to actual realized trades):

	GOOG	AAPL	MSFT	AMZN	INTC
N	44463	104459	312543	53016	321321
$2\hat{\alpha} = 2(\hat{H} + \frac{1}{2})$	0.674	0.712	0.724	0.703	0.794
$\hat{H}$	-0.163	-0.144	-.138	-0.149	-0.103

For intraday data, volume is of course highly time-inhomogenous (typically U-shaped - busier at the open and the close and quieter at lunchtime), so we should use a deterministic time-change of the F process in (1) if we are modelling over this time-scale. Note the empirical plots for signed order flow in Figures 1 and 2 in [MORS26] show signed trading volume over a time-scale of months/years, which avoids this intraday seasonality issue but also means we have far less sample points to estimate H.

1.7 Controlling liquidity

In the setup above, we have no control over the depth of liquidity, since the kernel $G(t)$ which controls the amount of transient price impact is endogenously determined via the f function in (1). To resolve this inflexibility, we can make the natural assumption that $P_t = \eta\tilde{M}_t + (1-\eta)\int_0^t G(t-s)dF_s$ for $\eta \in [0,1]$ where $\tilde{M}_t$ is an additional martingale, so η close to 1 implies greater liquidity and vice versa. The $\tilde{M}$ component accounts for changes in P due to news/external shocks (see also discussion on noise and informed traders in the introduction of [MB25]).

1.8 Using a mixed kernel

If we augment f to a function of the form $f(t) = \nu e^{-\theta t} t^{H-\frac{1}{2}} + \nu_2 e^{-\theta_2 t} t^{H_2 - \frac{1}{2}}$, then we find that

$$h(t) \quad = \quad \frac{1}{\nu\Gamma(\alpha)} \sum_{n=0}^{\infty} (-c)^n \sum_{k=0}^{\infty} \frac{(\beta n)_k}{k!}\, \delta^k\, \theta^{-(\mu_n + k)}\, \frac{\gamma(\mu_n + k, \theta t)}{\Gamma(\mu_n + k)}$$

where $\beta = \frac{1}{2} + H_2$, $c = \frac{\nu_2\Gamma(\beta)}{\nu\Gamma(\alpha)}$, $\delta = \theta - \theta_2$, $\mu_n = \beta n - \alpha(n+1)$, $\gamma(a,z) = \int_0^z s^{a-1}e^{-s}ds$ is the analytic continuation of the lower incomplete Gamma function and $(a)_k$ denotes the Pochhammer symbol. We can then use two (possibly different) f functions to drive $F^{\pm}$, and if $H_2 \in (0, \frac{1}{2})$, P_t will have an additional standard (i.e. non-hyper) rough component, which may be more realistic. We can also relax the assumption that M and M^- are independent to give greater control over the skewness of P_t.

1.9 Tempered long memory/multiscale behaviour

If we want $(F_t^{\pm})_{t\geq 0}$ to have longer memory/multiscale behaviour (as reported empirically in [MORS26]), we can replace f with e.g. $f(t) = (\frac{1}{c_1 t^{H-\frac{1}{2}}} + \frac{1}{c_2} t^{\gamma})^{-1}$ with $H \in (-\frac{1}{2}, 0)$, $\gamma > 1$ and $c_1, c_2 > 0$ (note f is still in L^1).

1.10 Power-law decay for the acf of order flow signs

We can also consider a discrete-time formulation where signed order $F_n = \nu B_n^H$ for $n = 1, 2, ...$, where B^H is an fBM with $H \in (\frac{1}{2}, 1)$, and let $Z_n = B_n^H - B_{n-1}^H \sim N(0,1)$ so $(Z_n)_{n=1}^{\infty}$ is fractional Gaussian noise (fGN) with $Z_n \sim N(0,1)$ for all n, then

$$\begin{aligned} \rho(n) \;:=\; \mathbb{E}(Z_{k+n} Z_k) &= \mathbb{E}((B_{k+n}^H - B_{k+n-1}^H)(B_k^H - B_{k-1}^H)) \\ &= \tfrac{1}{2}((n+1)^{2H} - 2n^{2H} + (n-1)^{2H}) \;\sim\; const. \times n^{2H-2} \qquad (n \to \infty) \end{aligned}$$

and we see that Z is stationary. For two correlated $N(0,1)$ random variables X, Y with $\mathbb{E}(XY) = \rho$ it can be shown that $\mathbb{E}(\mathrm{sgn}(X)\mathrm{sgn}(Y)) = \frac{2}{\pi} \arcsin \rho$. Let $\varepsilon_n = \mathrm{sgn}(Z_n)$. Then

$$C(n) \;=\; \mathbb{E}(\varepsilon_k \varepsilon_{k+n}) \;=\; \frac{2}{\pi} \arcsin \mathbb{E}(Z_k Z_{k+n}) \;=\; \frac{2}{\pi} \arcsin \mathbb{E}(Z_0 Z_n)\,.$$

But $\arcsin x = x + O(x^3)$ as $x \to 0$, so

$$\mathbb{E}(\varepsilon_k \varepsilon_{k+n}) \;\sim\; \frac{2}{\pi} \mathbb{E}(Z_k Z_{k+n}) \;\sim\; const. \times n^{2H-2}$$

as $|n| \to \infty$, so we have power-law decay for the a.c.f. of the signs of the order flow increments, similar to the metaorder models in [LMF05] and [MB25].

1.11 Bid-ask spreads, exogenous choice of impact kernel, and non-semimartingale price dynamics

We can also include a proportional transaction cost $\varepsilon P_t > 0$ per share bought (i.e. a bid-ask spread). If we then also abandon the (questionable) assumption that $P_t = \kappa \lim_{u\to\infty} \mathbb{E}(F_u^+ - F_u^- \,|\mathcal{G}_t)$, and instead exogenously choose the price impact kernel G and assume that e.g. $P_t = P_0 + \int_0^{t\wedge\tau_\delta} G(t-s) dF_s$ for some deterministic function G where $\tau_\delta = \inf\{t : P_t = \delta\}$ and $P_0 > \delta > 0$ with $\delta \ll 1$; then if P has the conditional full support property, the model is arbitrage-free even if P is not a semimartingale (see e.g. Guasoni et al.[GRS08] for details). Note also that the trading costs due to G are quadratic.

1.12 Automated Market Makers

Let x and y denote the amounts of two different coins in a constant product market maker liquidity pool. The basic Uniswap v1 protocol imposes that $xy = \kappa^2$ where κ is known as the *depth* of the pool, which is constant unless liquidity is added or removed from the pool by liquidity providers (LPs). If a Liquidity taker (LT) sells Δy coins to the pool, then the amount of X coins leaving the pool is $\frac{\kappa^2}{y} - \frac{\kappa^2}{y+\Delta y}$, and dividing this by Δy, and letting $\Delta y \to 0$, we see that the exchange rate for an infinitesimal amount of Y is $z = \frac{\kappa^2}{y^2} = \frac{xy}{y^2} = \frac{x}{y}$. We can then assume that the amount of Y coins in the pool at time t is $Y_t = Y_0 + F_t^+ - F_t^-$ until Y hits zero, so $Z_t = \frac{\kappa^2}{Y_t^2}$. In practice, there are also proportional transaction costs of $\varepsilon = .003$, so as in subsection 1.11, if Z satisfies the conditional full support property, the model is arbitrage-free (note we are not assuming existence of an additional traditional LOB-based exchange like Binance here).

1.13 Signed order flow as a persistent Gaussian process

Let $\bar{F}(t) = \int_0^t f(s) ds$ where $f(.)$ is defined as in subsection 1.1 with $H \in (-\frac{1}{4}, 0)$, and we now assume the signed order flow F follows the Gaussian Volterra process:

$$F_t \;=\; \int_0^t \bar{F}(t-s) dW_s \;=\; \bar{F}(t-s) W_s|_{s=0}^{s=t} + \int_0^t f(t-s) W_s ds \;=\; \int_0^t f(t-s) W_s ds$$

where W is a standard Brownian motion. Then $\bar{F}(t) = O(t^{H+\frac{1}{2}}) = O(t^{H+1-\frac{1}{2}})$ as $t \to 0$, i.e. F has Hölder continuity $H + 1 - \varepsilon \in (\frac{1}{2}, 1)$, and

$$\mathbb{E}(F_u | \mathcal{F}_t^W) \;=\; \int_0^t f(u-s) W_s ds + W_t \int_t^u f(u-s) ds$$

and $\lim_{u\to\infty} \mathbb{E}(F_u|\mathcal{F}_t^W) = c_{H,\nu,\theta} W_t$ and $W_t = \int_0^t h(t-s) dF_s$ (interpreted as a Young integral), so the model is plausible but leads to boring Bachelier dynamics for the price process P_t (it does have the advantage that we can use MLE or GMM methods to estimate H and θ). Or we can replace W with another martingale (e.g. a process with rough volatility).

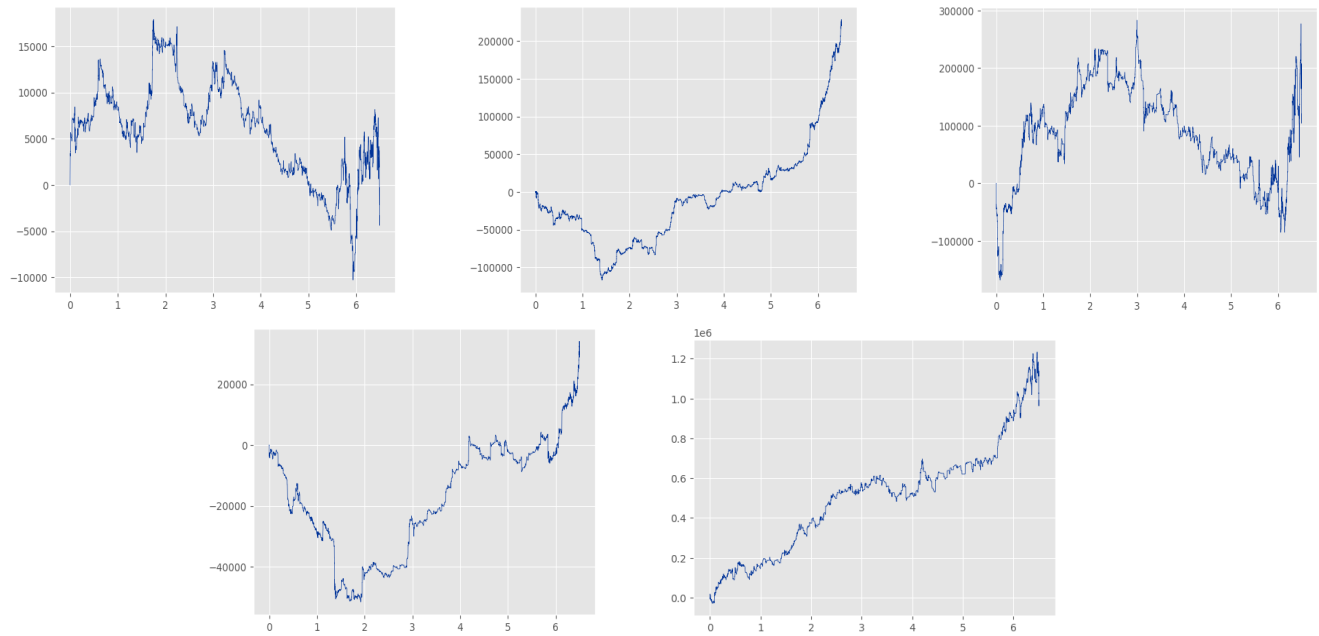

Figure 1: Signed order flow over a 6.5hr trading day for GOOG, AAPL, MSFT, AMZN and INTC

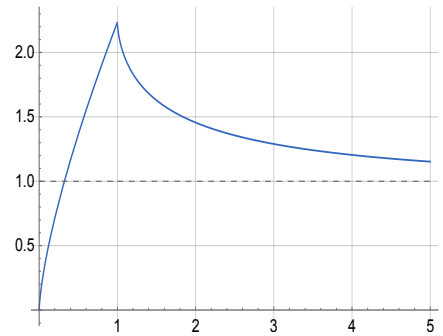

Figure 2: Concave price impact of an exogenous metaorder executed at constant trading speed 1 over $[0, t_0]$ with $H = -0.2$, $\theta = 0.1$, $\kappa = 1$ and $t_0 = 1$. The blue line asymptotes to the constant level $MI(\infty) = \kappa t_0 = 1$ in this case (grey line) as $t \to \infty$ which represents the asymptotic permanent price impact.

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A Appendix A

By Dambis–Dubins–Schwarz, there exists a Brownian motion B (on a possibly enlarged probability space) such that $M_t = B_{F_t^+}$. Now set $G_t := \sup_{0\le r\le t}|g(r)|$, $K_t := \int_0^t |f(r)|dr > 0$, and for $n \ge 1$, define the level hitting time $\tau_n := \inf\{s \ge 0 : F_s^+ \ge n\}$.

Since F^+ is continuous and nondecreasing, on $\{\tau_n \le t\}$ we have $F^+_{\tau_n} = n$ and $F^+_r \le n$ for all $0 \le r \le \tau_n$. Fix $n \ge 1$ and work on the event $\{\tau_n \le t\}$. For $0 \le r \le \tau_n$ we have $|B_{F_r^+}| \le \sup_{0\le u\le n}|B_u|$. Setting $s = \tau_n$ and taking absolute values we see that

$$\begin{aligned} n \;=\; g(\tau_n) + \int_0^{\tau_n} f(\tau_n - r)\, B_{F_r^+}\, dr \;&\le\; |g(\tau_n)| + \int_0^{\tau_n} |f(\tau_n - r)|\, |B_{F_r^+}|\, dr \\ &\le\; G_t + (\int_0^{\tau_n} |f(\tau_n - r)|\, dr) \sup_{0\le u\le n} |B_u| \\ &=\; G_t + (\int_0^{\tau_n} |f(s)|\, ds) \sup_{0\le u\le n} |B_u| \;\le\; G_t + K_t \sup_{0\le u\le n} |B_u|. \end{aligned}$$

Hence for $n > G_t$, on $\{\tau_n \le t\}$ we have $\{\sup_{0\le u\le n}|B_u| \ge \frac{n-G_t}{K_t}\}$. Since $F_t^+ \ge n$ iff $\tau_n \le t$, we obtain

$$\mathbb{P}(F_t^+ \ge n) \;\le\; \mathbb{P}(\sup_{0\le u\le n} |B_u| \ge \frac{n - G_t}{K_t}), \qquad n > G_t.$$

By the standard reflection principle bound $\mathbb{P}(\sup_{0\le u\le n}|B_u| \ge a) \le 4\exp(-\frac{a^2}{2n})$, for $a > 0$, so for $n > G_t$,

$$\mathbb{P}(F_t^+ \ge n) \;\le\; 4\exp(-\frac{(n - G_t)^2}{2nK_t^2}).$$

In particular, if $n \ge 4G_t$ then $(n - G_t)^2/n \ge \frac{1}{2}n$, hence

$$\mathbb{P}(F_t^+ \ge n) \;\le\; 4e^{-\frac{n}{4K_t^2}}, \qquad n \ge 4G_t. \tag{A-1}$$

Since $F_t^+ \ge 0$, we also know that $\mathbb{E}(F_t^+) = \int_0^\infty \mathbb{P}(F_t^+ \ge x)\, dx$. Now set $m := \lceil 4G_t \rceil$. Then

$$\mathbb{E}(F_t^+) \;\le\; m + \int_m^\infty \mathbb{P}(F_t^+ \ge x)\, dx \;=\; m + \sum_{n=m}^\infty \int_n^{n+1} \mathbb{P}(F_t^+ \ge x)\, dx \;\le\; m + \sum_{n=m}^\infty \mathbb{P}(F_t^+ \ge n). \tag{A-2}$$

Using (A-1) for all $n \ge m$,

$$\sum_{n=m}^\infty \mathbb{P}(F_t^+ \ge n) \;\le\; \sum_{n=m}^\infty 4\exp(-\frac{n}{4K_t^2}) \;<\; \infty. \tag{A-3}$$

Combining (A-2) and (A-3) yields $\mathbb{E}(F_t^+) < \infty$.